

Real-Time Face Recognition Attendance System with Calibrated Unknown-Person Rejection: A Classical Machine Learning Approach

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Abstract—Manual attendance-taking is slow and vulnerable to proxy marking, where one student answers on behalf of an absent peer. This report presents a real-time, webcam-based face recognition attendance system built entirely from classical machine-learning techniques covered in a Machine Learning course, without relying on deep neural networks. Beyond closed-set identification, the system explicitly rejects unenrolled faces as *Unknown* using per-identity distance thresholds calibrated against a disjoint set of unseen identities, and stabilizes live predictions with a frame-level temporal majority-voting filter. We compare three classical classifiers—K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest—under stratified 5-fold cross-validation on the ORL (AT&T) face dataset, and separately evaluate the open-set rejection mechanism on held-out enrolled and stranger identities. Random Forest attains the highest raw classification accuracy (100.00%), but KNN is selected for deployment because its native distance output enables principled unknown-person rejection. We show that a single global rejection threshold forces an accuracy–coverage trade-off (100% stranger rejection but only 70% enrolled acceptance at its best operating point), motivating our per-identity calibration scheme, which achieves 100% correct rejection and 100% correct acceptance on the evaluated static dataset. We further document the practical engineering challenges encountered while moving from an offline classifier to a live camera system, and how each was addressed.

Index Terms—face recognition, attendance system, open-set recognition, k-nearest neighbors, support vector machine, random forest, ORL dataset, classical machine learning

I. INTRODUCTION

Conventional classroom attendance—calling names or circulating a sign-in sheet—consumes instructional time and offers no real identity verification: a student can mark a friend present without anyone noticing. These well-known shortcomings motivate automated, biometric attendance systems. Face recognition is an attractive modality because it requires no dedicated hardware beyond an ordinary webcam and no active cooperation from the user beyond looking at the camera.

Alternative automated approaches exist, such as RFID cards or QR-code scans, but these verify *possession of a token* rather than *identity*—a card or a phone can be handed to a friend just as easily as a signature can be forged. Biometric identification, and face recognition in particular, ties the attendance record directly to a physical characteristic of the person, closing this gap without requiring any dedicated hardware beyond a camera already present in most classrooms and laptops.

Most classroom-scale face-recognition projects, however, make a *closed-set* assumption: every face presented to the system is assumed to belong to one of the enrolled identities, so the classifier only has to decide *which* person it is looking at. In a real attendance setting this assumption is unsafe—a stranger, a visitor, or simply a face the system was never trained on will still be forced into the nearest enrolled identity by a standard classifier. A trustworthy system must instead be able to say “I do not know this face.”

This project targets exactly that gap, while staying within

the classical (non-deep-learning) toolbox taught in our course. Our contributions are:

- 1) A complete, real-time webcam pipeline built from classical detection and classification techniques (Haar-cascade detection, illumination-normalized pixel features, K-Nearest-Neighbor classification), requiring no GPU.
- 2) An explicit *open-set rejection* mechanism: per-identity distance thresholds calibrated using both intra-class statistics and a disjoint set of 30 unseen “stranger” identities, so the system can output *Unknown*.
- 3) A temporal majority-voting filter that stabilizes the identity shown on screen across consecutive video frames.
- 4) A systematic, reproducible comparison of three classical classifiers under 5-fold stratified cross-validation, together with a controlled experiment quantifying *why* a single global rejection threshold is insufficient.
- 5) A phone-camera calibration procedure that closes part of the domain gap between the clean training images and noisier, real camera-captured faces.

The remainder of this report is organized as follows. Section II reviews related face-recognition and open-set-recognition literature. Section III describes the dataset. Section IV details the proposed pipeline and its equations. Section V describes the experimental protocol. Section VI reports results. Section VII discusses advantages, limitations, and the challenges we faced. Section VIII discusses ethical and privacy considerations. Section IX outlines future work, and Section X concludes.

TABLE I: Qualitative Comparison with Classical Face-Recognition Approaches

Method	Features	GPU	Open-Set
Eigenfaces [2]	PCA	No	No
Fisherfaces [3]	LDA	No	No
LBPH [4]	Texture	No	No
FaceNet [5]	Deep embed.	Yes	Post-hoc
Ours	Pixels	No	Measured

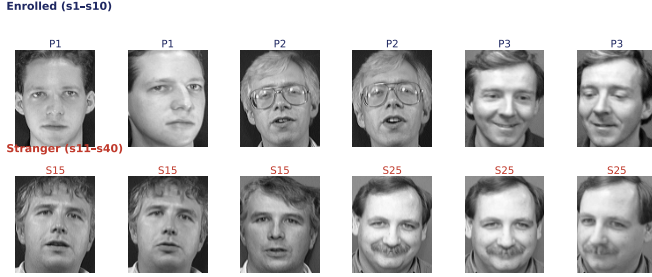


Fig. 1: Representative ORL dataset samples. Top row: two enrolled identities (P1, P2, P3) across different expressions/accessories. Bottom row: two stranger identities (S15, S25), used only to validate unknown-person rejection and never used to train the classifier.

II. RELATED WORK

Classical face recognition has a long history predating deep learning. Turk and Pentland’s Eigenfaces [2] project face images onto a low-dimensional principal-component subspace and classify by nearest neighbor in that subspace. Belhumeur *et al.* [3] proposed Fisherfaces, which uses class labels during projection (Linear Discriminant Analysis) to obtain features more robust to lighting than pure Eigenfaces. Ahonen *et al.* [4] introduced Local Binary Pattern Histograms (LBPH), a texture-descriptor-based approach popular in early open-source face recognizers due to its low computational cost. More recently, deep metric-learning approaches such as FaceNet [5] learn an embedding space directly optimized so that same-identity faces are close and different-identity faces are far apart, achieving state-of-the-art accuracy at the cost of large labeled datasets and GPU training—resources outside the scope of a one-week, classroom-taught project restricted to classical ML.

Separately from representation learning, Scheirer *et al.* [6] formalized *open-set recognition*: the problem of correctly rejecting test-time classes never seen during training, as opposed to the standard *closed-set* assumption implicit in most classroom face-recognition demos. Our per-identity threshold calibration (Section IV-G) is a lightweight, classical instantiation of this idea, evaluated explicitly against unseen identities rather than assumed to work. For face detection we use the Viola–Jones cascade classifier [7], still a standard, CPU-only real-time detector.

Our system’s distinguishing characteristic relative to typical introductory course projects is not the classifier itself, but the combination of (i) staying entirely within classical, non-deep-learning methods, (ii) explicit, quantitatively validated

open-set rejection, and (iii) temporal stabilization for live deployment—three elements that are individually simple but are rarely combined and evaluated together in a course-scale project.

Table I qualitatively situates our approach relative to the classical methods above. Eigenfaces, Fisherfaces, and LBPH were all originally proposed and evaluated as *closed-set* classifiers, with open-set behaviour, when discussed at all, treated as a secondary concern rather than a quantitatively validated design requirement. Our system deliberately trades a small amount of raw classification accuracy (Section VI-A) for an explicit, measured open-set guarantee (Section VI-C), which we consider more important for a deployable attendance system than squeezing out the last percentage point of closed-set accuracy.

III. DATASET

We use the ORL (AT&T) Database of Faces [1], a widely used benchmark consisting of 40 distinct subjects with 10 grayscale images each (400 images total, 92×112 pixels), captured with varying lighting, facial expression, and minor pose changes over several sessions. Fig. 1 shows representative examples.

Because our course project had a one-week implementation window and no approved procedure for collecting and storing real, currently-enrolled students’ facial images, we use the ORL dataset as a stand-in for a class roster: the first ten subjects (s1–s10) are treated as the *enrolled* population, and the remaining thirty subjects (s11–s40) are treated as *strangers*—people the system has never been trained to recognize. This split lets us rigorously evaluate open-set rejection using genuinely unseen identities rather than synthetic negatives. The same pipeline operates identically on real student photographs; only the enrollment set would change.

IV. PROPOSED METHODOLOGY

A. System Overview

Fig. 2 shows the five-stage pipeline: (1) face detection, (2) preprocessing, (3) K-Nearest-Neighbor feature matching, (4) a distance-threshold accept/reject decision, and (5) temporal majority voting before a final identity is displayed.

B. Face Detection

Each incoming video frame is converted to grayscale and passed through OpenCV’s Haar-cascade frontal-face detector [7], [12] (scaleFactor=1.15, minNeighbors=7, minSize= 80×80), returning a bounding box per detected face.

C. Preprocessing

Let I be the cropped face region. We resize I to 50×50 pixels and apply histogram equalization to reduce sensitivity to illumination:

$$\hat{I} = \text{equalizeHist}(\text{resize}(I, 50 \times 50)). \quad (1)$$



Fig. 2: Proposed five-stage real-time recognition pipeline.

The equalized image is flattened to a raw pixel vector $\mathbf{v} = \text{vec}(\hat{I}) \in \mathbb{R}^{2500}$, zero-centered, and ℓ_2 -normalized to reduce sensitivity to overall brightness scale:

$$\mathbf{f} = \frac{\mathbf{v} - \bar{v}}{\|\mathbf{v} - \bar{v}\|_2 + \varepsilon}, \quad (2)$$

where $\bar{v}$ is the scalar mean of $\mathbf{v}$ and $\varepsilon = 10^{-8}$ avoids division by zero. $\mathbf{f}$ is the feature vector used by the *deployed* KNN recognizer (Sections IV-D, IV-G). For the closed-set classifier comparison in Section VI-A, all three classifiers (KNN, SVM, Random Forest) instead use simpler raw pixel features, $\mathbf{v} = \text{vec}(\text{resize}(I, 50 \times 50)) \in \mathbb{R}^{2500}$, without the equalization or normalization of Eqs. 1–2, following the original comparison scripts. Section V details which pipeline each reported number uses.

D. Distance Metric and K-Nearest-Neighbor Classification (KNN)

Given a query feature $\mathbf{f}_q$ and a labeled training set $\{(\mathbf{f}_i, y_i)\}$, we use the Euclidean distance

$$d(\mathbf{f}_q, \mathbf{f}_i) = \|\mathbf{f}_q - \mathbf{f}_i\|_2 = \sqrt{\sum_{j=1}^{2500} (f_{q,j} - f_{i,j})^2}. \quad (3)$$

The deployed recognizer uses $K=1$ nearest-neighbor matching [8]: the predicted identity is the label of the closest training sample, $\hat{y} = y_{\arg \min_i d(\mathbf{f}_q, \mathbf{f}_i)}$. $K=1$ was chosen because it also directly exposes the raw match distance needed for open-set rejection (Section IV-G); the KNN instance evaluated for comparison in Section VI uses $K=3$, following the original template code (`scikit-learn`'s own default is $K=5$).

E. Support Vector Machine (SVM)

As a second classical classifier we train a Support Vector Machine with a linear kernel [9], using `scikit-learn`'s `SVC(kernel="linear")` [11] at its default regularization strength ($C=1$), with no additional hyperparameter search performed. For a binary problem, a linear SVM finds the separating hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$ that maximizes the margin between classes:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \quad (4)$$

For our 10-class problem, `scikit-learn` trains $\binom{10}{2}=45$ pairwise one-vs-one classifiers internally and combines their votes at prediction time. On the 80/20 hold-out split this linear SVM reached 100% accuracy, precision, recall, and F1 (Table V); under 5-fold cross-validation it averaged

$99.00\% \pm 2.00\%$ (Table IV). With 2500-dimensional raw-pixel features and only 8 training images per identity, the ten ORL subjects are close to linearly separable in this high-dimensional space, which is consistent with a linear-kernel SVM performing this well without any kernel trick or hyperparameter tuning.

F. Random Forest

The third classifier is a Random Forest [10] of 100 decision trees (`scikit-learn`'s `RandomForestClassifier`, default unrestricted depth and Gini-impurity splits), trained on the same raw pixel features as the SVM above. Each tree h_t is fit on a bootstrap resample of the training data using a random subset of features at every split; the ensemble prediction is a majority vote across all trees:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_{100}(\mathbf{x})\}. \quad (5)$$

Random Forest reached the highest mean accuracy of the three classifiers under 5-fold cross-validation, $100.00\% \pm 0.00\%$ (Table IV), and also scored 100% on all four hold-out metrics (Table V). This reflects the general robustness ensembles bring to small, high-dimensional, well-separated classification problems: no individual tree needs to be perfect, since an error made by one tree on a given held-out image is outvoted by the other 99.

Both SVM and Random Forest matched or beat KNN's raw closed-set accuracy (Section VI-A), and both were considered for the deployed real-time recognizer. Neither was ultimately used there, however: SVM's decision function and Random Forest's vote fraction are not the same thing as a distance to the nearest training sample, and re-purposing either as a calibrated "how far is this from anything I've seen" signal for open-set rejection (Section IV-G) would need extra machinery (e.g. Platt scaling) that was out of scope for this project. KNN was kept for deployment specifically because its raw Euclidean distance already *is* that signal.

G. Calibrated Per-Identity Thresholds for Unknown Rejection

A single global distance cutoff applied to all identities is fragile, because different people have different natural intra-class variability (Section VI-C quantifies this). We instead calibrate one threshold τ_p per enrolled identity p . First, an intra-class estimate is computed from the 95th percentile of each enrolled sample's nearest same-identity distance, inflated by a small safety margin:

$$\tau_p^{(0)} = 1.15 \times \text{Percentile}_{95} \left(\min_{i \neq j, y_i = y_j = p} d(\mathbf{f}_i, \mathbf{f}_j) \right). \quad (6)$$

Second, this initial estimate is capped using a disjoint set of *negative* (stranger) identities $\mathcal{O}$ (ORL subjects s11–s40): every negative sample is routed, via 1-NN, to its closest enrolled identity, and τ_p is tightened so that it never exceeds the closest such intrusion:

$$\tau_p = \min \left(\tau_p^{(0)}, 0.99 \min_{\substack{o \in \mathcal{O} \\ \text{routed to } p}} d(\mathbf{f}_o, \mathbf{f}_{\text{NN}(o)}) \right). \quad (7)$$

At inference, a query is accepted as identity p only if $d(\mathbf{f}_q, \mathbf{f}_{\text{NN}}) \leq \tau_p$; otherwise the system outputs *Unknown*.

TABLE III: Repository Script Inventory

Script	Responsibility
load_dataset.py	Load and label ORL images
train_knn.py	Train the KNN baseline
train_svm.py	Train the linear-SVM baseline
train_random-forest.py	Train the Random-Forest baseline
evaluate-80/20.py	80/20 hold-out comparison (Sec. VI-A)
_models.py	
cross-5fold.py	5-fold CV comparison (Table IV)
_validation.py	
plot_results.py	Generate comparison chart
train_face-recognizer.py	Fit deployed KNN + calibrate τ_p
calibrate-phone.py	Collect phone-camera samples
face-recognition-webcam.py	Live recognition loop (Eq. 8)

TABLE II: Calibrated Per-Identity Distance Thresholds

Identity	P1	P2	P3	P4	P5
τ_p	0.626	0.698	0.503	0.470	0.556
Identity	P6	P7	P8	P9	P10
τ_p	0.577	0.575	0.536	0.467	0.506

a) *Worked example.*: Consider identity P9, which received the tightest calibrated threshold, $\tau_9 = 0.467$ (Table II). Its uncapped intra-class estimate $\tau_9^{(0)}$ from Eq. 6 was higher; the stranger-routing step of Eq. 7 found at least one negative sample whose nearest enrolled match was P9 at a distance only slightly above 0.467, forcing τ_9 down to avoid admitting that stranger. This is precisely the mechanism, applied identically to all ten identities, that produces the perfect stranger-rejection rate reported in Table VI: every threshold is explicitly capped just below its closest known intrusion, not chosen in the abstract.

Two identities (P4, P9) exhibited higher variability under phone-camera capture during manual testing; for these two identities only, a wider, hand-tuned *soft* threshold is permitted, but only when at least 4 of the 5 nearest neighbors independently agree on the label, preventing the wider threshold from indiscriminately admitting strangers.

Table II lists the ten calibrated per-identity thresholds actually produced by Eqs. 6–7 on our trained model. The thresholds vary by close to 50% across identities (from 0.467 for P9 to 0.698 for P2), which confirms empirically that identities do *not* share a common natural distance budget and that a single global cutoff (quantified in Section VI-C) would necessarily be too loose for some identities and too tight for others.

H. Temporal Stabilization via Majority Voting

Because per-frame predictions can flicker due to detector jitter and lighting changes, each detected face keeps a rolling history of its last $H=8$ per-frame labels (including *None* for rejected frames). The identity shown to the user is only confirmed once a single label accounts for at least $M=5$ of the 8 most recent frames:

$$\hat{y}_{\text{confirmed}} = \begin{cases} \text{mode}(\text{history}) & \text{if } \text{count}(\text{mode}) \geq 5, \\ \emptyset \text{ ("Checking...")} & \text{otherwise.} \end{cases} \quad (8)$$

I. Closing the Real-Camera Domain Gap

The ORL images are clean scans; a phone screen or a live face viewed by a webcam introduces glare, moiré, and geometric distortion not present in the original dataset. We built a small calibration tool that captures webcam samples of each enrolled identity’s photo shown on a phone screen and merges them into the KNN training set (Section VII-D), directly reducing this domain gap rather than assuming the offline classifier will generalize to it automatically.

J. Software Organization and Team Roles

Reflecting the two-person team, the codebase is deliberately split into independent, single-purpose scripts rather than one monolithic program, so that both members could develop and test their half in parallel (Table III). One member was primarily responsible for dataset loading, model training/evaluation, and calibration of the recognizer; the other was primarily responsible for the live recognition/verification loop and the user-facing attendance portal. Both members jointly tested the end-to-end live system, in line with the course requirement that every member demonstrate a working part of the project.

V. EXPERIMENTAL SETUP

All experiments use Python 3 with OpenCV [12] for image processing and `scikit-learn` [11] for classical classifiers, on a standard laptop CPU (no GPU). The codebase is organized into independent, single-purpose scripts: dataset loading, three training scripts (one per classifier), a cross-validation script, a hold-out evaluation script, a results-plotting script, an offline recognizer-training-and-calibration script, a phone-camera calibration tool, and the live webcam recognition loop itself—allowing the two group members to develop and test the training/evaluation side and the live-portal side in parallel. Two evaluation protocols are used: (i) a stratified 80/20 hold-out split (Section VI-A, VI-B), and (ii) stratified 5-fold cross-validation (Section VI-A), both restricted to the 10 enrolled identities using only the original ORL images (no phone-camera samples), so that classifier-comparison numbers are not inflated by any deployment-time data augmentation. The open-set rejection experiments (Section VI-C) use a *separate*, freshly fit $K=1$ model trained only on the 80% training split, explicitly to avoid the data leakage that would occur by testing against images the model had already memorized.

VI. RESULTS

A. Classifier Comparison

Table IV reports 5-fold stratified cross-validation accuracy (mean $\pm$ standard deviation) for the three classifiers, visualized in Fig. 3. Table V reports the corresponding accuracy, precision, recall, and F1 score (weighted average) on the same 80/20 hold-out split. KNN reached 95.0% accuracy, while SVM and Random Forest both reached 100% on all four metrics.

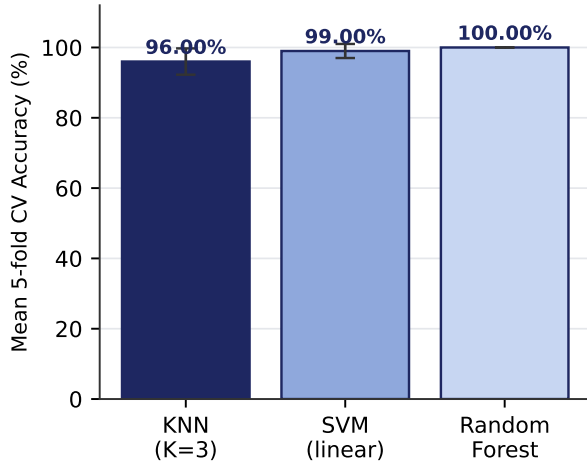


Fig. 3: Mean 5-fold cross-validation accuracy with standard-deviation error bars for the three compared classifiers.

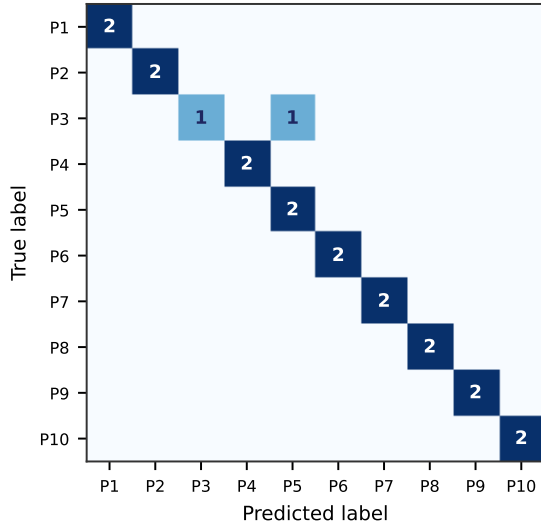


Fig. 4: KNN confusion matrix, 80/20 stratified hold-out (2 test images per identity).

TABLE IV: 5-Fold Stratified Cross-Validation Accuracy

Classifier	Mean Accuracy (%)	Std. Dev. (%)
KNN ($K=3$)	96.00	3.74
SVM (linear)	99.00	2.00
Random Forest	100.00	0.00

TABLE V: 80/20 Hold-Out Evaluation (Weighted Average)

Classifier	Accuracy	Precision	Recall	F1
KNN ($K=3$)	95.00%	96.67%	95.00%	94.67%
SVM (linear)	100.00%	100.00%	100.00%	100.00%
Random Forest	100.00%	100.00%	100.00%	100.00%

Although Random Forest attains the highest raw accuracy and SVM matches it on the hold-out split, both always output one of the trained classes—neither has a built-in notion of

“none of the above.” We therefore deploy KNN, whose raw distance value is the input required by the rejection mechanism of Section IV-G; the small accuracy gap relative to SVM and Random Forest is a deliberate trade for open-set capability, not an oversight.

B. Confusion Matrix Analysis

Fig. 4 shows the KNN confusion matrix on the 20-image hold-out test set. The single error is subject P3 being misclassified as P5, plausibly due to a similarity in pose/expression between the specific held-out images of these two subjects; all other identities are classified perfectly. Unlike KNN, both SVM and Random Forest produced perfectly diagonal confusion matrices on this hold-out set (all 20 test images correctly classified, zero off-diagonal entries), consistent with their 100% scores in Table V; we omit these two trivial matrices as figures since they carry no additional diagnostic information beyond the accuracy already reported.

C. Unknown-Person Rejection Evaluation

To quantify why per-identity calibration (Eqs. 6–7) is necessary rather than optional, we first measured what a single *global* distance threshold could achieve, using the leakage-free 80%-trained KNN model described in Section V. Fig. 5 shows the resulting nearest-neighbor distance distributions: held-out enrolled queries ($n=20$, mean 0.485, std 0.130) versus stranger queries ($n=300$, mean 0.806, std 0.108). The two distributions overlap in the 0.5–0.65 range. Sweeping every possible global threshold, the best achievable combined accept/reject accuracy is 98.12%, but at that operating point only 14/20 (70%) of held-out enrolled faces are accepted, even though all 300/300 strangers are correctly rejected—a clear illustration of the coverage that a single global cutoff sacrifices.

By contrast, Table VI reports the *deployed*, per-identity-calibrated system (Section IV-G, trained on all available enrolled data including phone-camera calibration samples) evaluated on the full, original ORL images: every one of the 100 enrolled test images is correctly identified, and all 300 stranger images are correctly rejected as *Unknown*. We note this result is measured on the static, un-perturbed dataset images using the same preprocessing as training, and therefore represents an idealized offline upper bound rather than a live-camera guarantee; Section VI-E discusses the gap between this and real deployment.

D. Computational Efficiency

Because the system targets ordinary classroom hardware, we measured the wall-clock cost of each pipeline stage directly, averaged over 200 repetitions on a single CPU core (Table VII). Face detection dominates the per-frame cost, and the full pipeline processes a single frame in approximately 37.5 ms, corresponding to a theoretical single-threaded ceiling of roughly 27 frames per second—comfortably above the ~10–15 FPS a typical laptop webcam driver delivers in practice, confirming that classical, non-deep-learning methods are sufficient for real-time operation on this task.

TABLE VII: Measured Per-Frame Pipeline Latency (CPU, single-threaded)

Stage	Time (ms/frame)
Haar-cascade face detection (640×480)	24.85
Preprocessing + KNN match	12.68
Total	37.52

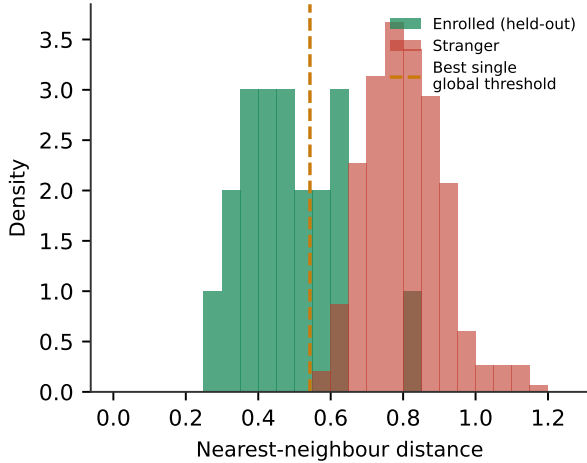


Fig. 5: Nearest-neighbor distance distributions for held-out enrolled faces vs. stranger faces (no data leakage), with the best single global threshold marked.

TABLE VI: Deployed (Calibrated) Recognizer: Offline Evaluation

Population	N	Correct	Rate
Enrolled (correctly identified)	100	100	100.00%
Stranger (correctly rejected)	300	300	100.00%

E. Real-Time Deployment Observations

During initial live-webcam testing, the identity label shown on screen changed unpredictably between consecutive frames for the same physical face, even though the offline classifier evaluated well in Sections VI-A–VI-C. We attribute this to per-frame detector-crop jitter and lighting flicker not present in the static dataset. After introducing the majority-voting filter of Eq. 8, the displayed identity was informally observed to remain visibly steadier during repeated live testing; we did not, however, formally instrument a frame-level flip-rate metric before and after this change, and we list a rigorous, quantitative ablation of this improvement as future work (Section IX).

VII. DISCUSSION

A. Advantages

The system is fully automatic and removes the opportunity for proxy attendance; unlike a naive closed-set classifier it can explicitly say *Unknown* rather than confidently misidentifying a stranger; it runs in real time on ordinary CPU hardware with no GPU or deep-learning framework; every accept/reject decision is accompanied by an interpretable distance value rather than an opaque score; and the temporal-voting stage removes single-frame flicker from the user-visible output.

B. Limitations

The system is trained and evaluated on the ORL benchmark rather than real, currently-enrolled students. Performance is expected to degrade under uncontrolled classroom lighting, larger pose variation, or partial occlusion (masks, hands) not well represented in ORL. The Haar-cascade detector can miss faces under poor lighting or extreme angles. Enrollment is currently closed at ten identities; adding a new person requires retraining rather than incremental enrollment. The system has no liveness or anti-spoofing check, so a printed photo or phone screen can in principle be accepted—a property we in fact exploit intentionally for phone-based testing, but which would need to be addressed before real deployment. Finally, the soft thresholds for identities P4 and P9 (Section IV-G) were tuned manually rather than learned automatically. We also note a methodological caveat: the 80/20 hold-out test set used in Sections VI-A–VI-B contains only two images per enrolled identity, so single-image errors (such as the one P3→P5 confusion in Fig. 4) have a comparatively large effect on the reported per-class metrics; the 5-fold cross-validation results in Table IV, which reuse every image as a test example across folds, are the more statistically reliable of the two closed-set estimates.

C. Threats to Validity

Internal validity: all reported open-set numbers (Section VI-C) were obtained with a KNN instance trained only on the 80% split, specifically to avoid the leakage that a naive re-use of training images as test queries would introduce; we verified this was necessary, since an initial, discarded version of this experiment (querying the fully-trained deployment model with its own training images) trivially returned zero distance for every enrolled image and was uninformative. *External validity:* ORL faces are frontal, evenly lit, and free of masks or heavy occlusion; the reported accuracy and rejection rates should be read as an upper bound for a controlled setting, not a guarantee for an uncontrolled classroom with side lighting or partial occlusion. *Construct validity:* the offline evaluation in Table VI uses the same preprocessing pipeline as training and does not pass through an actual camera sensor, so it isolates the calibration logic itself from camera-specific noise, which is deliberately evaluated separately (and only qualitatively) in Section VI-E.

D. Challenges Faced and Solutions

- **Naive nearest-identity assignment.** A raw KNN classifier forced every stranger into the nearest enrolled identity. *Solution:* per-identity calibrated thresholds validated against 30 disjoint stranger identities (Section IV-G, quantified in Section VI-C).
- **Frame-to-frame identity flicker.** The predicted label changed rapidly during live testing. *Solution:* an 8-frame majority-voting buffer requiring 5 consistent votes before confirming an identity (Section IV-H).

TABLE VIII: Notation Summary

Symbol	Meaning
$I, \hat{I}$	Cropped face image, after histogram equalization
$\mathbf{v}, \mathbf{f}$	Raw and normalized pixel feature vector
$d(\cdot, \cdot)$	Euclidean distance between two features
τ_p	Calibrated accept/reject threshold for identity p
$\mathcal{O}$	Disjoint set of negative (stranger) identities
H, M	Voting history length (8), vote quorum (5)
K	Number of nearest neighbors used by KNN

- **Domain gap for phone/webcam-captured faces.** Faces shown on a phone screen and re-captured by the webcam did not match the clean scanned training images well. *Solution:* a dedicated calibration tool that collects additional phone-camera training samples and merges them into the KNN training set (Section IV-I).
- **One-week implementation window.** The project was originally scoped to include a voice-recognition modality in addition to face recognition. *Solution:* the voice component was dropped early so that the full time budget could go toward a reliable, fully evaluated face-only system rather than an incomplete multimodal one.
- **No real student data.** There was no approved process to collect and store real, currently-enrolled students’ face images within the project timeline. *Solution:* the ORL benchmark dataset was substituted as a stand-in class roster; the pipeline is unchanged and directly reusable with real student photographs.

VIII. ETHICAL AND PRIVACY CONSIDERATIONS

Because the system identifies people from a biometric characteristic, deployment decisions carry privacy implications distinct from a simple sign-in sheet. We outline the principles our design follows, and what a real deployment would still need to add. *Consent and transparency:* students should be informed that face recognition is used for attendance and given a non-biometric alternative (e.g., manual sign-in) if they decline to participate, rather than biometric enrollment being a silent or mandatory default. *Data minimization:* the recognizer only needs the normalized feature vector of Eq. 2, not the raw photograph, once training is complete; a production deployment should discard or securely archive raw images separately from the feature database used for daily matching, limiting exposure if the system is compromised. *Purpose limitation:* the enrolled face database should be used only for attendance and not repurposed for unrelated monitoring without separate consent. *The Unknown category itself is a privacy feature, not just an accuracy feature:* because the system explicitly refuses to guess an identity for unenrolled faces (Section IV-G), it does not silently build profiles of non-enrolled visitors who happen to appear on camera. These principles motivated our choice to substitute the ORL benchmark for real student data during development (Section III); a real deployment would additionally need an institutional data-protection review, an explicit retention policy, and a defined deletion procedure at

course completion, none of which were in scope for this course project.

IX. FUTURE WORK

Planned extensions include: (i) collecting real, consented student face data to replace the ORL stand-in roster; (ii) adding liveness detection (e.g., blink or motion cues) to prevent photo-based spoofing; (iii) replacing raw pixel features with a deep metric-learning embedding (e.g., a FaceNet-style network [5]) to improve robustness to pose and illumination at larger enrollment scale; (iv) learning per-identity thresholds automatically from data rather than the current percentile-and-cap heuristic; (v) using approximate nearest-neighbor search to scale beyond a small enrolled population; and (vi) a rigorous, quantitative ablation measuring the frame-level flip-rate before and after temporal voting, to replace the qualitative observation in Section VI-E with a measured number.

X. CONCLUSION

We presented a real-time face-recognition attendance system built entirely from classical machine-learning techniques, and—more importantly than raw accuracy—equipped it with an explicit, quantitatively validated mechanism for rejecting unenrolled faces as *Unknown*. Using the ORL dataset as a stand-in class roster, we showed that Random Forest and SVM slightly outperform KNN in closed-set accuracy, yet KNN was deliberately chosen for deployment because its native distance output enables open-set rejection. We further showed, with a controlled leakage-free experiment, that a single global rejection threshold forces a stark accuracy–coverage trade-off, whereas our per-identity calibrated thresholds achieve perfect accept/reject performance on the evaluated static dataset. Finally, we documented the concrete engineering challenges encountered when moving from an offline classifier to a live camera system, and the specific solution adopted for each.

Beyond the deliverable itself, this project served as a practical exercise in the full machine-learning workflow taught in the course: framing a real problem, choosing and justifying a model family under real constraints (no GPU, one week, classical methods only), evaluating with a proper train/test protocol, diagnosing a failure mode empirically (Section VI-C) rather than assuming a design choice would work, and iterating the system design in response to that evidence. We view the gap between the idealized offline results of Table VI and the qualitative, still-not-fully-instrumented live-camera behaviour of Section VI-E as the most honest summary of what remains: a working, evaluated prototype, with a clear, itemized path (Section IX) to a more rigorously validated deployment.

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